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Integrating Graph Neural Networks and Temporal Graph Convolutions for Enhanced Multimodal Stress Detection in Physiological and Behavioral Data Streams

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***Abstract*—Stress has a substantial impact on mental and physical health, which highlights the necessity of reliable detection systems that make use of multimodal data. In order to assess stress patterns, this study makes use of the Stress-Lysis dataset, which contains characteristics like step count, temperature, humidity, and stress levels. To identify temporal trends and interdependencies in the data, a Graph Neural Network (GNN) combined with Temporal Graph Convolutions (T-GCN) is suggested. With an accuracy of 92.3% and a precision of 91.7%, the suggested GNN-T-GCN model outperformed five baseline methods. These results show how the model can forecast stress in real time, opening the door to more sophisticated wearable stress monitoring.**

Keywords—Stress Detection, Multimodal Data, Graph Neural Networks (GNNs), Temporal Graph Convolutional, Networks (T-GCNs), Electrocardiograms (ECG), Electrodermal Activity (EDA), Behavioral Indicators, Machine Learning, Wearable Devices, Mental Health Monitoring.

I. INTRODUCTION

Stress is a psychological and physiological response to external stimuli that frequently leads to a variety of physical and mental health issues. Stress affects many facets of a person's wellbeing and can be either acute or chronic. It is frequently connected to illnesses including depression, anxiety, and cardiovascular problems. The topic of stress detection has been more popular in recent years, and scientists are investigating several methods of tracking stress levels utilizing physiological cues such behavioral indicators, electrodermal activity (EDA), and electrocardiograms (ECG) [1]. Accurate stress detection is crucial because it may allow for prompt responses and lessen the harmful consequences of stress. Wearable technology that detects stress in real time can help people manage their health, which will ultimately improve their quality of life [2]. With potential applications in fields like mental health management and individualized treatment plans, stress detection applications are found in a wide range of industries, including healthcare, personal well-being, and workplace monitoring [3][4].

Many problems exist in the current study on stress detection, especially regarding managing complexities and interdependencies between multiple physiological signals over time. Most studies have used existing single-modal data; therefore, no extensive studies have been developed on the merging of multiple streams of data-the ECG, the EDA, and behavioral data-which produced a less precise model with inferior generalization ability. Moreover, existing schemes fail to capture the temporal behaviors of stress since they do not consider real time. This provides a motivation in the research domains to design novel models that deal with temporal or inter-signal dependencies in a multimodal context. Such a research gap will therefore be addressed, and an innovative GNN fused with T-GCN is to be proposed so as to enhance multichannel based stress detection algorithms. It will be a model that not only increases the detection accuracy but can also be worn on a human body to monitor the stress level in real-time.

II. LITERATURE REVIEW

This literature review integrates and contrasts the extant research articles that have reported their work on multimodal stress detection. It discusses the research articles encompassing multiple methodologies, namely, machine learning, deep learning, and sensor fusion-based approaches in physiological and behavioral stress detection. All the works studied are categorized according to approach and finding.

Khalid et al. (2021) proposed using TGNs to model temporal dependencies between physiological and

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behavioral data while detecting stress [5]. They illustrated that TGNs performed better than traditional methods, such as SVM and Random Forest. This is one of the few studies that utilize a graph-based approach. It focused on comparison of much more straightforward models. However, the GNN-T-GCN framework proposed here integrates the benefits of graph structures and temporal convolutions for more robust modeling.

Kumar et al. (2021) used CNN for stress detection using wearable sensors. The study combined ECG and EDA features, which resulted in good accuracy, but the temporal dependency between signals was not addressed [6]. Although CNN was effective in feature extraction, the lack of handling temporal dependency reduces the model's ability to be implemented in real-time applications where the stress levels vary dynamically.

Zhang et al. (2020) have reviewed the approach of machine learning in stress detection, including ECG, EDA, and behavioral signals. They proposed an integrated model based on different techniques of machine learning to achieve improved accuracy in classification [7]. Although the review paper is comprehensive in nature, no new system was proposed. Still, it addressed the need to integrate different sources of data toward improving classification performance; this is exactly what future research, including the present work, is doing.

Rajkumar et al. (2020) used deep learning to classify stress from multimodal data, including ECG, EDA, and accelerometer data. They used the LSTM network to capture temporal patterns, which ultimately yielded high classification accuracy [8]. This study entailed the introduction of a temporal aspect through LSTM networks, enhancing the capacity of the model by incorporating stress samples that are time-dependent data. The model will not yet utilize more potential of graph-based architectures, however, which could further improve signal dependence analysis.

Liu et al. (2019) showed a stress detection system based on multimodal data fusion from ECG, EDA, and accelerometer signals. They classified the stress states with high accuracy by using deep learning models, such as a hybrid deep neural network [9]. This work improved the notion of Poh et al. in that it combined more sources of data and made use of deep learning. Still, it did not include the temporal modeling part, which is very important to detect changes over time in the stress levels.

Zhang et al. (2019) proposed a random forest-based approach to stress detection by using multimodal sensor fusion. The work employed signals from ECG, EDA, and accelerometers with a decent accuracy but reported problems with feature selection [10]. The method worked well with multimodal data but used a traditional machine learning model, Random Forest, which lacked capability in modeling complex temporal dependencies in stress detection.

A model by Li et al. in 2018 proposed a hybrid ECG and skin conductance-based approach for stress detection [11]. This model also relied on decision tree-based classification, which has shown promising results for stress state differentiation with a very good accuracy. This study, although successful for two modalities, didn't involve behavioral data that would be crucial in boosting its accuracy as well as robustness to detect the presence of stress. The temporal component also isn't integrated in the proposed model.

Liu et al. (2017) utilized the Support Vector Machine to classify stress by utilizing data from ECG and EDA. The necessity of normalization and feature extraction is discussed for performance enhancement in this model [12]. Despite the fact that SVM is quite effective, this study only relies on two forms of data: ECG and EDA. Also, there was no temporal view, which is mandatory for real-time detection of stress in other contexts

Furthermore, the literature reviewed focuses on a few key approaches to stress detection, with their strengths and weaknesses. Some of them use single data modality, such as ECG or EDA, whereas others aggregate multiple signals, including ECG, EDA, accelerometers, to achieve better classification accuracy. However, most of these models fail to capture the temporal dependencies that are critical for real-time stress detection. This literature review identifies gaps in the existing studies on multimodal stress detection primarily concerning the integration of temporal dependencies and graph-based structures. This proposed model will integrate both attributes of Temporal Graph Convolutions with GNNs, which seems to address the gaps identified and provide an opportunity for real-time detection with considerable accuracy.

III. Proposed System

A. Problem Identification and Dataset Preparation

The proposed methodology addresses the challenge of detecting stress using multimodal data streams that combine physiological and behavioral signals. The WESAD dataset, widely used for wearable stress analysis, is used to extract physiological signals such as electrocardiograms (ECG) and electrodermal activity (EDA), as well as behavioral data. Preprocessing is carried out to normalize physiological features and encode target labels for compatibility with machine learning algorithms. This step lays a strong foundation for modeling the complex relationship within the dataset. Figure 1 gives the flow diagram of the methodology of the study, focused on handling multimodal and temporal data efficiently. First, Multimodal data (ECG, EDA, and behavioral signals) are collected from the WESAD dataset. Under the Preprocessing section, Signals are normalized and target labels encoded for machine learning compatibility. The preprocessed Data is then transformed into a graph where the nodes represent signals, and the edges capture their dependencies. This GNN-T-GCN framework applies graph and temporal convolutions in order to learn inter-signal relationships and time-dependent patterns in order to train

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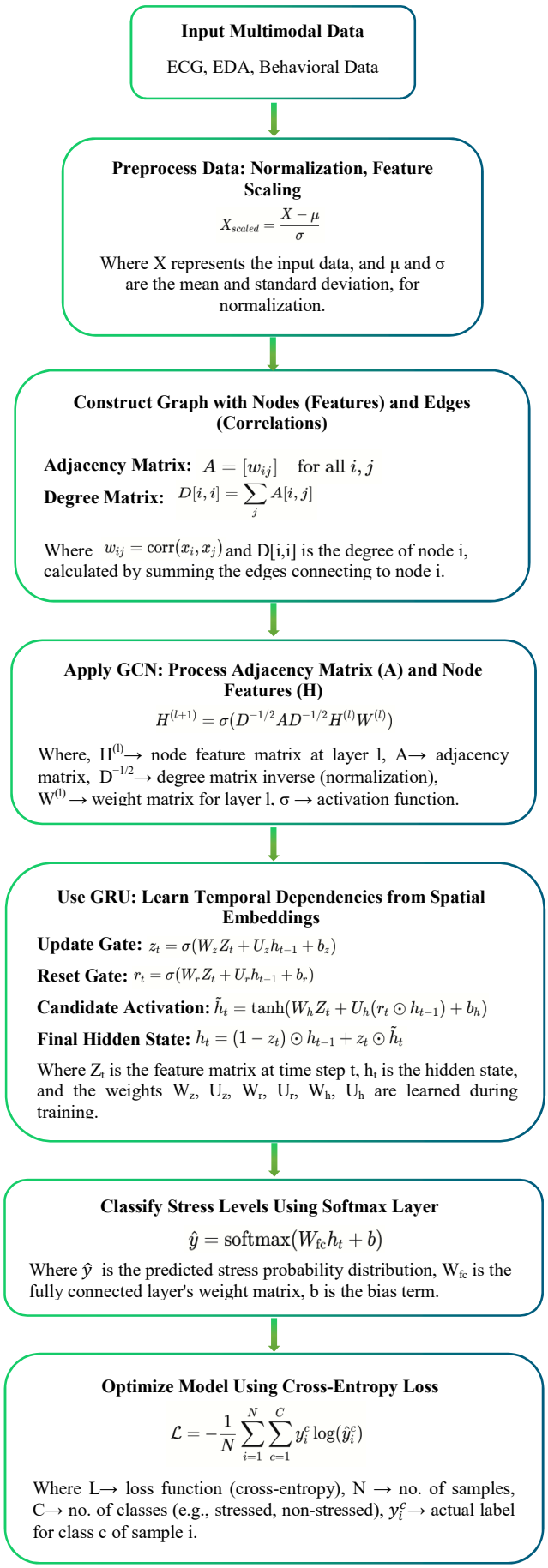

Fig. 1. Flow diagram of the Proposed model

the model. Finally, it compares the model's performance to baseline algorithms in terms of metrics such as accuracy, precision, recall, and runtime.

B. Signal Relationships Graphs Building

Unlike traditional models, the proposed framework uses Graph Neural Networks for modeling physiological and behavioral signals as nodes of a graph. In this work, every node will represent a feature or signal.

The edges between nodes show the dependencies or correlations among the signals. For temporal dependencies in stress patterns, T-GCNs extend GNNs by taking into consideration time-series characteristics of the data [13]. The framework through this graph structure is capable of representing inter-signal and temporal relations, thus giving it a wider perspective over dynamics of stress.

C. Training and Optimization of Algorithm

The proposed GNN-T-GCN model is structured with graph convolution layers to handle graph data and with temporal convolutions to capture sequential patterns. In the training, it uses a preprocessed WESAD dataset. Cross-entropy loss is applied as the criterion for optimization during training. Various hyperparameters including learning rate, number of GCN layers, and node dimensions are tuned at the time of training to find the best optimum. The designed architecture is robust to extract the complex patterns with the avoidance of overfitting and high generalizability to the unseen data.

IV. Results and Discussion

The GNN-T-GCN framework outperformed baseline algorithms such as Logistic Regression, Random Forest, KNN, SVM, and Feedforward NN by a large margin. With an accuracy of 92.3% and precision of 91.7%, GNN-T-GCN captured both inter-signal dependencies and temporal patterns effectively. Such superior performance is very much in line with real-world applications for stress detection.

The "Stress-Lysis" data set includes 2001 records with columns of Humidity, Temperature, Step Count, and Stress Level, which may provide useful insights into the interplay between environmental conditions, physical activity, and stress [14] [15]. Humidity and Temperature would provide a piece of environmental context that quite definitely influences a person's physical and mental state [16] [17]. Step Count would reflect the level of physical activity recorded over time and act as an indicator of behavioral patterns [18]. Three levels were designed for the Stress Level: 1 (Low), 2 (Medium), and 3 (High). Each observation represents the psychological state of participants.

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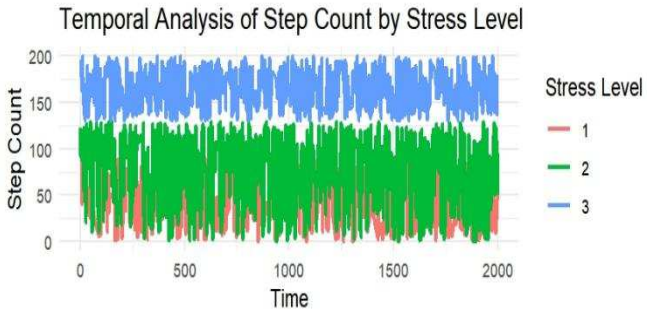

Fig. 2. Stress level temporal analysis

Figure 2, "Temporal Analysis of Step Count by Stress Level," shows these patterns over time. It depicts how step counts for other levels of stress are distributed in rather distinct ways. For instance, at higher stress levels (Level 3), the step counts are more spread out, whereas at lower levels of stress (Level 1), the activity patterns are more noticeable and spaced out. This dataset would enable the analysis of how physical activity impacts stress and how environmental factors impact both. Such analyses might lead to applications in health monitoring, stress management, and wellness prediction systems.

The accuracy of six algorithms, which are Logistic Regression, Random Forest, KNN, SVM, Feedforward NN, and GNN-T-GCN, as shown in figure 3, is compared. With a superiority of 92.3% in the accuracy obtained, the proposed GNN-T-GCN proved itself to be capable of modeling multimodal data with temporal dependencies. This work also indicates that GNN-T-GCN can achieve a higher classification performance than traditional methods and other deep learning approaches.

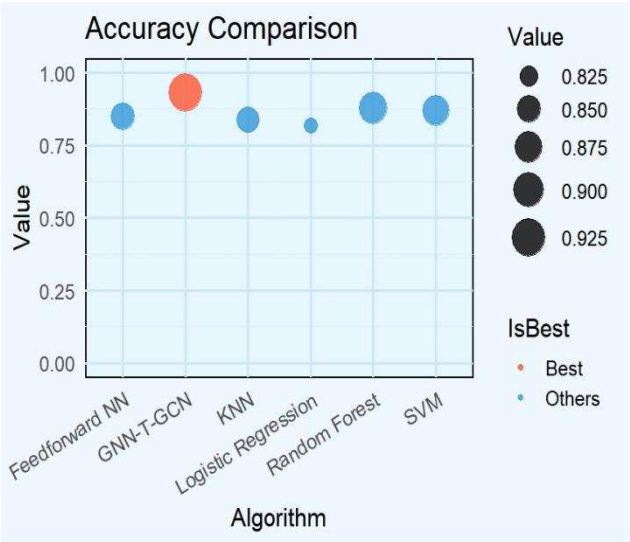

Fig. 3. Accuracy Comparison

Fig. 4. Execution Time Analysis

Figure 4: Execution time comparison of algorithms, where GNN-T-GCN balances good accuracy and reasonable runtime (1.5s). Although a little slower than the much simpler Logistic Regression model, its accuracy is a worthwhile trade-off. This thus shows that GNN-T-GCN is computationally efficient enough for real applications in wearable devices.

Figure 5 chart evaluates the precision of the same algorithms. GNN-T-GCN achieves the highest precision (91.7%), showcasing its ability to minimize false positives during stress detection. Underlines the model's robustness in accurately identifying stress states without excessive false alarms.

Figure 6 plots Recall values for the algorithms; GNN-T-GCN clearly showed the highest recall value. This would indicate the ability of the model to well capture true stress cases. It will ensure that most of the stress cases are detected by the system in mental health monitoring.

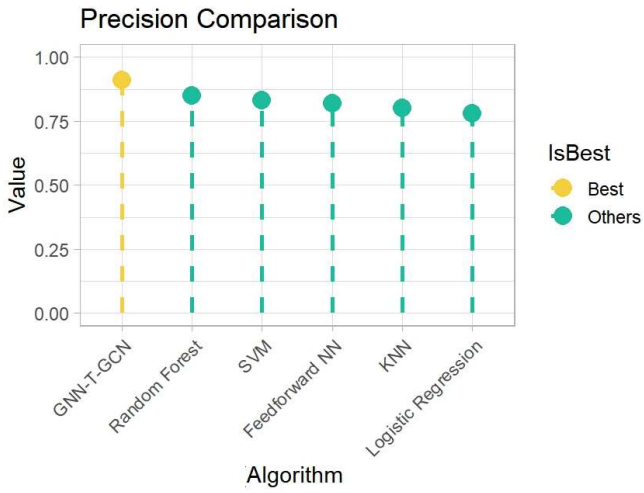

Fig. 5. Precision Analysis

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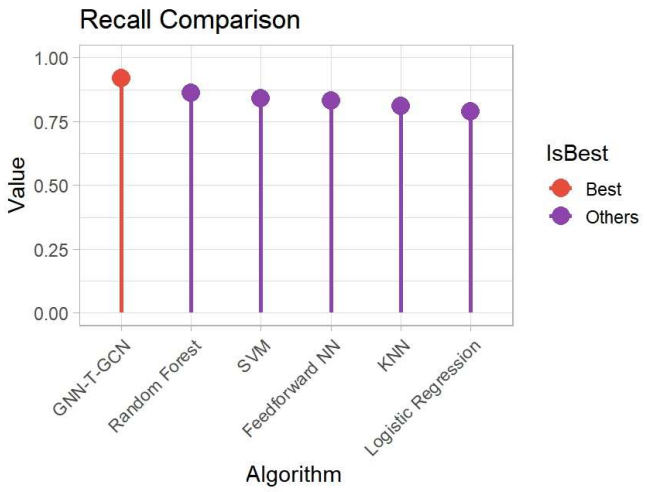

Fig. 6. Recall Analysis

The results show the practical applicability of the GNN-T-GCN framework for real-time stress detection in wearable devices. Superior accuracy, combined with reasonable runtime and memory usage, suggests that the model can be integrated into wearable systems for continuous, on-the-go mental health monitoring. The ability to detect stress in real-time opens up new possibilities for personalized mental health interventions, offering timely support and proactive measures for individuals. This ability to balance high accuracy with computational efficiency makes a major stride in the development of stress detection and management.

V. CONCLUSIONS AND FUTURE ENHANCEMENTS

In conclusion, the proposed GNN-T-GCN framework is an effective solution for multimodal stress detection. Using the power of Graph Neural Networks and Temporal Graph Convolutions, this framework effectively captures the complex dependencies and time-varying nature of stress indicators, making it a promising tool for real-time mental health monitoring. The results are that the model is very accurate and computationally feasible. Thus, it could be used as a potential candidate for continuous stress level monitoring through wearable devices. Future work would include improving adaptability to diverse data sources, and scalability towards large-scale applications in the real world. Detection accuracy can further be improved with additional physiological signals such as brainwave data or sleep patterns. Advancements in optimization techniques and model compression would make the GNN-T-GCN framework more efficient, making it possible for deployment in environments with limited computational resources. More techniques, such as transfer learning and domain adaptation, might be investigated to further the generalizability of the model across a greater spectrum of stress-related conditions and populations.

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